

Transnational Access (TNA) Report

ATRIUM Computer Vision in Archaeology Training School

Participant: Árpád Balogh

Host institution: Institute of Archaeology of the Czech Academy of Sciences, Brno (ARÚB), via the Archaeological Information System of the Czech Republic (AIS CR)

Dates: 14–18 September 2026, Brno, Czech Republic

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1. Biographical Data and Research Background

I am Árpád Balogh, a GIS specialist at the National Archaeological Institute of the Hungarian National Museum and a PhD student in Earth Sciences at the Budapest University of Technology and Economics. My primary research focuses on the management of spatial databases, the structuring of cultural heritage data according to FAIR principles, and the Python-based automation of remote-sensing analyses alongside broader geoinformatic workflows.

2. Host Institution and Details of the Stay

As a recipient of a Transnational Access (TNA) grant, I participated in the ATRIUM Computer Vision in Archaeology Training School, held in Brno, Czech Republic, from 14 to 18 September 2026. The school was organised as a transnational access provision of the ATRIUM Project, funded under the Horizon Europe research and innovation programme, and hosted by the Archaeological Information System of the Czech Republic (AIS CR) research infrastructure at the Institute of Archaeology of the Czech Academy of Sciences, Brno. The school was co-organised by the MAIA COST Action.

Teaching was delivered by a team of five instructors, supported by the AIS CR administrative and technical staff. The cohort brought together many participants from across Europe and beyond, working in archaeology and related disciplines — which made both the taught sessions and the informal exchanges between participants unusually rich.

3. Work Performed and Training Content

The programme was structured around five real-world archaeological case studies—artefacts, microscopic use-wear and micromorphology, satellite imagery, rock art, and coins—introduced progressively to cover foundational and advanced computer vision workflows.

Practical exercises were conducted locally using Python/Jupyter environments and via Google Colab for GPU-intensive tasks.

The training began with a conceptual introduction to computer vision and hands-on Python programming, covering zero-shot image classification using CLIP and fundamental OpenCV/Matplotlib pipelines for image segmentation and measurement. Following this, the focus shifted to dataset construction and annotation, utilizing interchange formats like COCO and YOLO, and exploring assisted polygon annotation with CVAT and SAM 2 to address common issues like class imbalance and inconsistent metadata. Additionally, Tuesday afternoon was set aside for participants to present their own ongoing research. This proved highly valuable, as it allowed us to learn about each other's work and fostered a wider discussion regarding the diverse computer-vision problems being tackled across the cohort.

Mid-week sessions delved into microscopy image analysis for use-wear and micromorphology, employing techniques such as histogram matching, watershed segmentation, and TensorFlow data augmentation to standardise and quantify heterogeneous image sets. Remote sensing applications were also a major component; participants built a complete pipeline to detect archaeological cropmarks in Sentinel-2 multispectral imagery using NDVI, Frangi ridge filters, and Random Forest classifiers with hard-negative mining, producing GIS-ready vector data.

Later sessions addressed small-object instance segmentation for Scandinavian rock art. Participants worked with 3D scan visualisations enhanced by local relief models, fine-tuning a YOLO instance-segmentation model to detect cupmarks. To preserve the scale of these small features, the workflow utilized tiled inference via the SAHI library, ultimately exporting the predicted masks as georeferenced vector layers. The practical training then concluded with a high-level overview of complex, cutting-edge vision-language models applied to coin datasets, illustrating the broader horizon of emerging AI technologies in the field. The week wrapped up with a general group discussion on the application of machine learning in archaeology.

4. Contribution of the TNA Grant to Ongoing Research

The breadth of the programme maps unusually well onto my own work. The dataset-annotation formats and workflows introduced on Tuesday are directly relevant to how spatial and image collections are structured and published in line with FAIR principles, and to the controlled-vocabulary problems I regularly encounter when integrating heterogeneous source data.

More directly still, Thursday morning's satellite remote-sensing pipeline is architecturally very close to the remote-sensing automation strand of my PhD research, and I intend to reuse this pipeline structure, adapted to my own study areas, as a starting point for feature-detection work there.

The small-object, tiled-inference approach used for cupmark detection in the rock art session is similarly transferable to any dense or small-feature detection problem in my own heritage-mapping work, including the 3D point-cloud classification strand of my thesis, where the same logic of stacking complementary features and validating iteratively against known error modes applies just as well. Beyond specific pipelines, the broadened toolkit built up across the sessions strengthens the geoinformatic automation scripts I maintain day to day. Finally, the training school connected me with an international, genuinely cross-disciplinary cohort, which I expect to be valuable for future collaboration and for keeping up with methodological developments in computational archaeology.

5. Acknowledgements

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